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Contents

1	Bootstrap from OIS rates	4
2	Most Liquid Contract	6
3	C-spread	8
3.a	Single time-series of C-spread	10
4	Z-index	11
5	3-months risk-free rate	13
6	Time-series plot and analysis	14
7	Johansen procedure for cointegration analysis	17

8 ECM: Error Correction Model	19
9 Robustness checks	21
9.a EWMA Volatility	21
9.b Phase III and Phase IV	23
9.c ROLL-OVER OPEN INTEREST	26
9.d ROLL-OVER one month	28
9.e ROLL-OVER one week	28
10 Quantile Regression	30
11 EUA synthetic repo	33
12 Matlab Code	35

Abstract

The big aim of this project is to solve the Puzzle of Carbon Allowance spread. Particularly the word *puzzle* stays for an anomaly observed in the EUA market: the C-spread, i.e. the spread over the risk-free rate in the relation between future and spot prices, is constantly positive. This fact has no intuitive meaning, since there is no cost of storing allowances and at the same time there is no benefit to keep the spot EUA (as there is obviously for commodities for example).

Solving this market's inefficiency can help European Union to limit greenhouse gases emissions. Following Azzone, Baviera, and Manzoni (2024) we get statistical evidence that the C-spread is cointegrated with both credit spread and risk-free interest rate. In particular, we show that, on the long run, the main driver of the C-spread is the Z index, i.e. a proxy for the credit spread of the EU-ETS players.

The European Union Emissions Trading System (EU ETS) is the main instrument adopted by the European Union to achieve CO₂ reduction targets in key industrial sectors. The mechanism is of the cap-and-trade type, meaning it sets an overall maximum cap on emissions allowed in the covered sectors across the European territory (cap). This corresponds to an equivalent number of "allowances" (1 tonne CO₂eq. = 1 allowance) that can be bought and sold on a dedicated market (trade). Every industrial operator active in the sectors covered by the scheme must offset their actual emissions on an annual basis with a corresponding amount of allowances.

1 Bootstrap from OIS rates

We were asked to bootstrap the risk-free interest rates from OIS rates following the methodology described in Baviera and Cassaro (2014). We started by importing and preprocessing the quoted Euro Overnight Index Swap (OIS) rates, filling up missing values via flat interpolation.

OIS contracts with maturity shorter than one year impose an exchange at a unique date of the difference between a rate, the R^{OIS} , fixed in t_0 , and a floating rate fixed at maturity, depending on the EONIA rates, having the following formula:

$$R(t_s, t_e) = \frac{\prod_{k=t_s}^{t_e-1} [1 + \delta(t_k^E, t_{k+1}^E) E(t_k^E)] - 1}{\delta(t_s, t_e)}$$

Working with EONIA as discounting curve, we can compute the discounting factors as the expected values of the stochastic discount $D(t_0, t_e)$, therefore obtaining the following formula (see Baviera and Cassaro (2014) for details):

$$P^D(t_0, t_e) = \frac{1}{1 + \delta(t_0, t_e) R^{OIS}(t_0, t_e)}$$

On the other hand, OIS contracts with longer maturities are not bullet instruments: the first payment date isn't fixed, but it changes in order to allow constant yearly payments for the following ones. The discounts' curve therefore requires different calculations:

$$P^D(t_0, t_i) = \frac{1 - R^{OIS}(t_0, t_i) \sum_{k=1}^{i-1} \delta_k P^D(t_0, t_k)}{1 + \delta_i R^{OIS}(t_0, t_i)},$$

Once we got the discounts' curve, we could easily obtain the zero rates:

$$r(t_0, t_e) = -\frac{\log(P^D(t_0, t_e))}{\delta(t_0, t_e)}$$

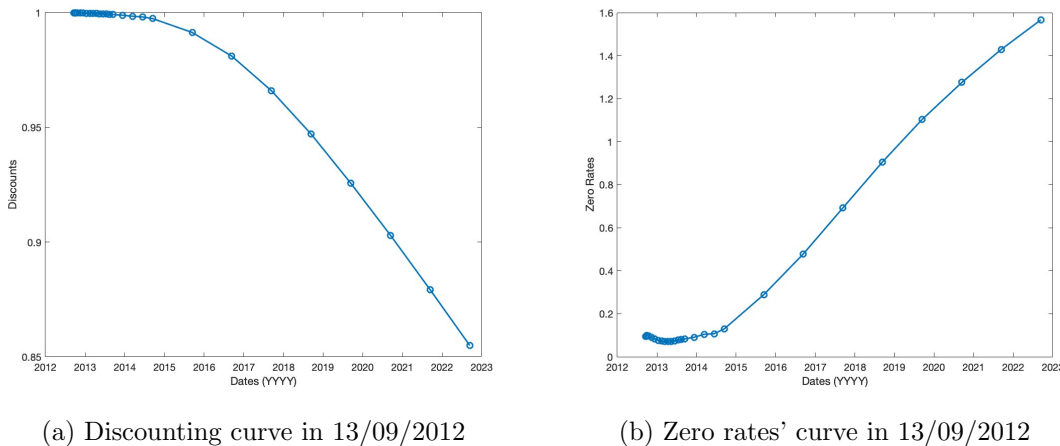


Figure 1: Bootstrap results over 10 years

It is worth to consider this aspect: *why did we use OIS rates and did not rely on the classical bootstrap procedure with depos, futures and IR swaps?* As mentioned in Baviera and Cassaro (2014) and in Hull and A. White (2012), the OIS discounting procedure has become, after the credit crisis started in 2007, the new standard for practitioners. We try to briefly show the principal reasons:

1. Overnight rates are a better indicator of short-time credit risk, so the zero rates that we get are a very precise representation of risk-free rates
2. Following the crisis, the interbank curve is no longer unique since large basis spreads have appeared between different Euribor tenors and consequently the standard bootstrap can no longer be implemented (Baviera and Cassaro (2014))
3. In modern financial markets many transactions, especially those connected to derivatives, are collateralized. When a derivative contract is collateralized, the rates applicable to the associated cash flows are generally overnight rates, reflecting the cost of short-term financing. Since OIS are built on overnight rates, they provide a more accurate benchmark for discounting collateralized cash flows.

2 Most Liquid Contract

Our objective for this qualitative analysis was to prove that the EUA December Front contract is the most liquid. We firstly defined the most liquid contract as the one with the highest volume, we then used a box-plot to have a qualitative comparison.

At our disposal we had the volumes sold for the Front contract in March, June, September and December, after retrieving them from the dataset we were given, we built four box-plots.

These box-plots were built using a log-scale. Considering that especially for the June and September Front contracts there were many days in which the traded volumes were equal to 0, we summed one to all of the volumes of the series (in this way we managed to compute the logarithm of 0) and obtained a good box-plot. One more thing to notice is that we did not use any kind of Roll-Over rule, we took the entirety of the Front contracts, from an expiry to the one of the following year.

We can clearly see that the December Front contract is the most liquid among the four.

We then compared the possible types of December contracts: Front, Next and Next Next. We applied the same procedure as before, with the only difference that in order to create the series of volumes we only focused on December future contracts, taking each time the correct volumes to add to the series.

The box-plots we obtained are the following:

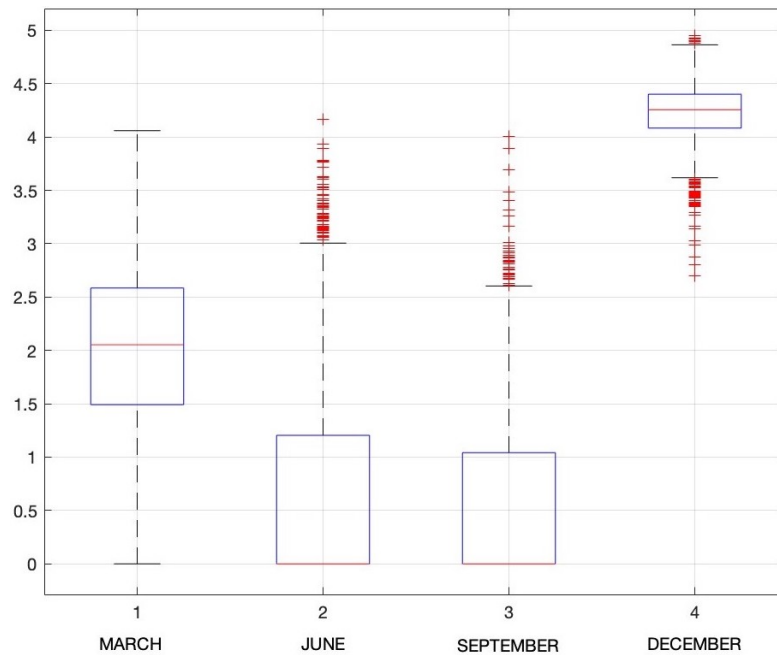


Figure 2: Boxplot for Front future contracts' volumes in log-scale

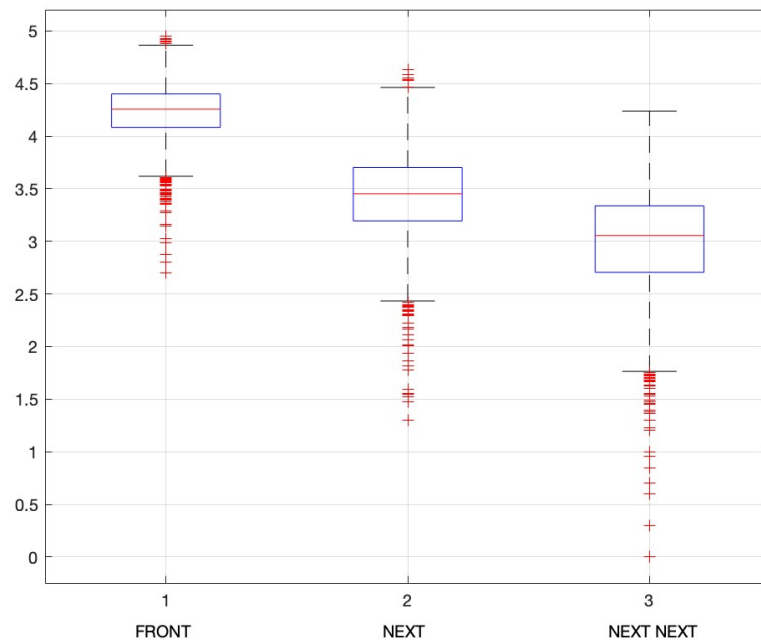


Figure 3: Boxplot for December future contracts' volumes in log-scale

Looking at these box-plots it is clear that the Front December contract is the most liquid between the three December future contracts. From this point on we focused our analysis mostly on the Front December contracts.

3 C-spread

The aim of this section was to compute the cost-of-carry spread over the risk-free rate (C-spread), both for the Front and the Next December future contract.

To do so we implied equation 2.1 of (Azzone, Baviera, and Manzoni 2024):

$$C_t = \frac{\log(\frac{F_{t,T}}{S_t})}{T-t} - r(t, T)$$

where

- T is the delivery date
- $F_{t,T}$ is the future price with delivery date in T : we retrieved them from the table regarding future prices
- S_t is the spot price EUA spot price: since we did not have them, we used the daily future prices as proxy
- $r(t, T)$ is the risk-free rate between t and T : we used the ones obtained from the bootstrap

Since not all the data were available for each date, we had to define a way to proceed in order to have the highest possible number of dates. What we decided to do was to find the common dates between the future Front prices and the daily future prices, this were the dates that we used for our calculations and for each one of them we obtained the value for the C-spread. Not all these dates were however present among the bootstrap's start dates. In order to solve this issue we created a complete set of bootstrap's start dates employing a flat extrapolation using the data from the previous date.

We then constructed the C-spread curve one year at a time. For each year, we identified the date on which the Front December contract commenced and the corresponding expiry. For each date belonging to this time interval, we found the daily future price (to use as proxy for S_t), the future Front price and lastly we needed to deduce the risk-free rate. The main issue was that for each date the distance from the maturity was different and we were almost never able to find it precisely in the data from the bootstrap. So, we linearly interpolated the value of the risk-free rates, for each date we were interested in.

At this point, we were able to compute the C-spread, for each date, since we had all the parameters needed by the previous formula.

The procedure we have just described is the one needed in order to compute the C-spread for the Front December contracts. The mechanism to obtain the C-spread derived from Next December contract is almost analogous. The only difference comes when having to consider the prices for the future contract and the time interval to imply. In this scenario, as the start date we considered the start date of the Front contracts of the previous year, whereas as the end date we used delivery

date again of the previous year's. It is important to notice that the maturity, instead, remains the same as the Front contract case. We then took the future prices accordingly to the time interval we have just defined.

After all these computations we managed to obtain two time-series for the C-spread, one based on Front December contracts and the other one based on Next December contracts.

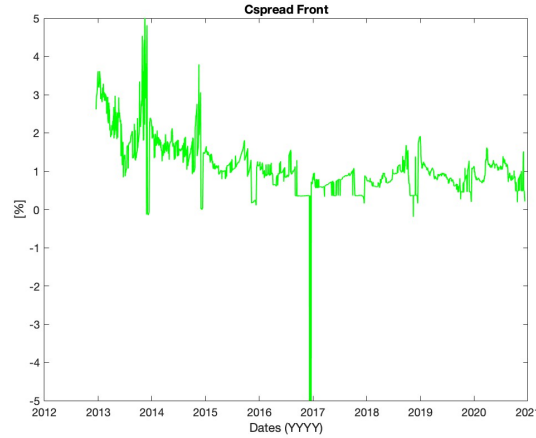


Figure 4: C-spread Front December in percentage

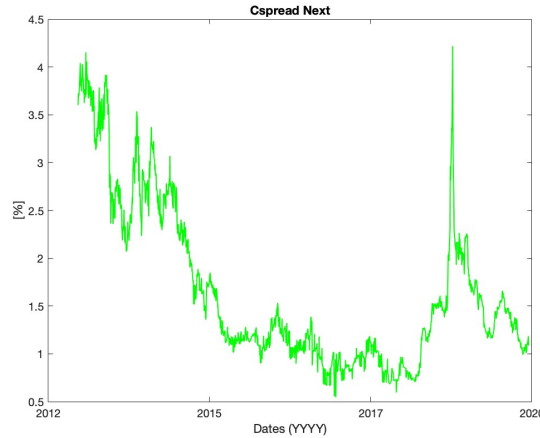


Figure 5: C-spread Next December in percentage

From these plots we can notice how there are some high and low peaks, especially in the C-spread Front. This behaviour is connected to the formula that we implied, since at the denominator you can find the year fraction between the delivery date and the date in which we are evaluating the C-spread, as the date progressively gets closer to the delivery, the denominator tends to zero, causing the peaks we are observing. Something worth to mention regards the very low value there is around the 16th of December 2016, this behaviour is a consequence of a combination of factors. First of all the date is extremely close to the delivery date for that year, consequently on this day we are experiencing a future price smaller than the daily future, meaning that the logarithm returns

a negative result, which divided by something very close to 0, tends to be even more negative. Apart from that, the two series have the same order of magnitude, they are however defined on two different time periods. We are considering Phase III, that ends at the end of 2021, this means that in the case of Front contracts we will reach the last delivery in 2021, whereas with the Next December contract we will have values that go up until the previous year.

3.a Single time-series of C-spread

The procedure we are now going to present is used to obtain a single time-series for the C-spread by combining the two time-series we have previously built.

In order to get just one time-series we used the following Roll-Over rule: starting from the first day of the Front December contract, we considered the values of the Front C-spread up until the 15th of November of the closest year. From that date up to the next 15th of November, we considered the future contract with maturity in the following year, thus taking the Next future until the same year maturity, and then the Front one until a new switch.

Looking at the C_t in 8 we can notice how this Roll-Over rule allowed us to remove the peaks from the time-series we had previously observed. Since we are not considering the last trading days of the Front contracts, which were the most critical, by substituting them with the Next December contracts the problematic points are not taken into account. As affirmed in Azzone, Baviera, and Manzoni (2024), this is the standard practice in EUA futures market, as operators not interested in entering into a spot EUA position in December, shortly before the maturity, rollover their exposure into the Next December contract.

4 Z-index

In Azzone, Baviera, and Manzoni (2024), it's stated that it's possible to construct a proxy of the average credit spread in the EU-ETS market, the Z-index, as the weighted average of the Z-spreads of the bonds emitted by the 12 European issuers having the highest Scope 1 carbon equivalent emissions. Of those 12 issuers, we considered only non-convertible fixed coupon bonds having notional higher than 500 millions.

Being the Z-spread the spread that would need to be added to the risk-free zero rate curve in order to have the discounted cash flows of the bond equal to its current market price, we needed to compute the NPV for each coupon bond.

For the missing dates in our zero-rates curve, we interpolated linearly and extrapolated flat, computing then the discount factors $B(t_0, t) = e^{-\delta(t_0, t)r(t_0, t)}$, that we utilized to obtain the Z-spreads. We computed them for each bond proceeding issuer by issuer, calculating then each player's Z-spread as the mean of the Z-spreads of its emitted bonds weighted by their issued amounts.

At this point we conducted a simple data analysis on the issuers' Z-spreads and we noticed a weird behaviour of the ENI's curve. Looking at the dataset, the bond "XS0877820422" resulted to be problematic, since its price was too high as the time tended to maturity. We decided not to use this particular bond in order to get a better shaped Z-index time-series. Moreover, for the purpose of deleting some big outliers values for Z_t , we decided to introduce a threshold of 0.075 for the time delta in Z-spread formula: when, near to maturity, delta was lower than this value, we manually set $Z = 0$ and didn't account this value in the final average.

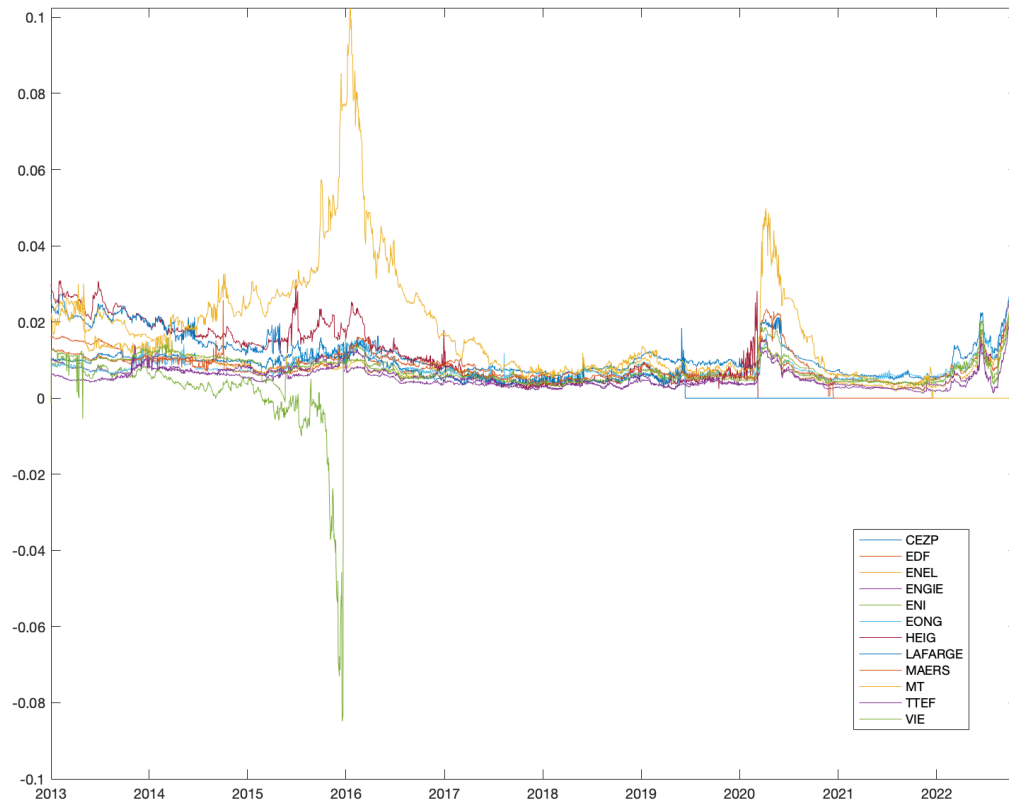


Figure 6: Z-spreads of the 12 issuers

We finally computed the Z-index Z_t as the average of the issuers' Z-spread (8).

5 3-months risk-free rate

After the Great Financial Crisis, as previously mentioned in section 1, it's standard practice to utilize Euro OIS swaps for the risk-free rates' curve bootstrap instead of traditional interbank rates such as the LIBOR.

Therefore, we only needed to construct the time-series of the risk-free rate before starting an in-depth econometric analysis to try to understand the relationships between C_t , Z_t and specifically r_t . We built the time-series of r_t just by considering for each date the corresponding bootstrapped 3-months zero rate, as specified by the following table.

	1w	2w	3w	1m	2m	3m	4m	5m
04/01/10	0,0032	0,0034	0,0035	0,0034	0,0037	0,0037	0,0038	0,0039
05/01/10	0,0033	0,0035	0,0035	0,0034	0,0036	0,0037	0,0038	0,004
06/01/10	0,0033	0,0035	0,0034	0,0034	0,0036	0,0037	0,0038	0,004
07/01/10	0,0033	0,0035	0,0034	0,0034	0,0036	0,0037	0,0038	0,004
08/01/10	0,0035	0,0035	0,0033	0,0034	0,0037	0,0037	0,0039	0,004
09/01/10	0,0036	0,0035	0,0034	0,0034	0,0037	0,0037	0,0038	0,0039
10/01/10	0,0036	0,0034	0,0033	0,0035	0,0037	0,0037	0,0038	0,004
11/01/10	0,0036	0,0036	0,0034	0,0034	0,0037	0,0037	0,0038	0,0039
12/01/10	0,0034	0,0035	0,0033	0,0034	0,0036	0,0036	0,0037	0,0038
13/01/10	0,0034	0,0033	0,0033	0,0034	0,0036	0,0036	0,0037	0,0038
14/01/10	0,0034	0,0033	0,0033	0,0034	0,0036	0,0036	0,0037	0,0038
15/01/10	0,0032	0,0031	0,0032	0,0033	0,0035	0,0035	0,0036	0,0037
16/01/10	0,0032	0,0031	0,0033	0,0033	0,0035	0,0036	0,0037	0,0038
17/01/10	0,0032	0,0031	0,0032	0,0033	0,0035	0,0036	0,0037	0,0038
18/01/10	0,0034	0,0031	0,0032	0,0033	0,0035	0,0035	0,0036	0,0038
19/01/10	0,0033	0,0031	0,0031	0,0033	0,0035	0,0036	0,0037	0,0038
20/01/10	0,0029	0,0031	0,0032	0,0033	0,0035	0,0036	0,0037	0,0038
21/01/10	0,0029	0,0031	0,0031	0,0033	0,0035	0,0036	0,0037	0,0038

Figure 7: Snapshot of our bootstrap table (first 8 columns and 18 rows)

6 Time-series plot and analysis

After having computed the three time-series for the variables of interest, we plotted them in the Phase III time window.

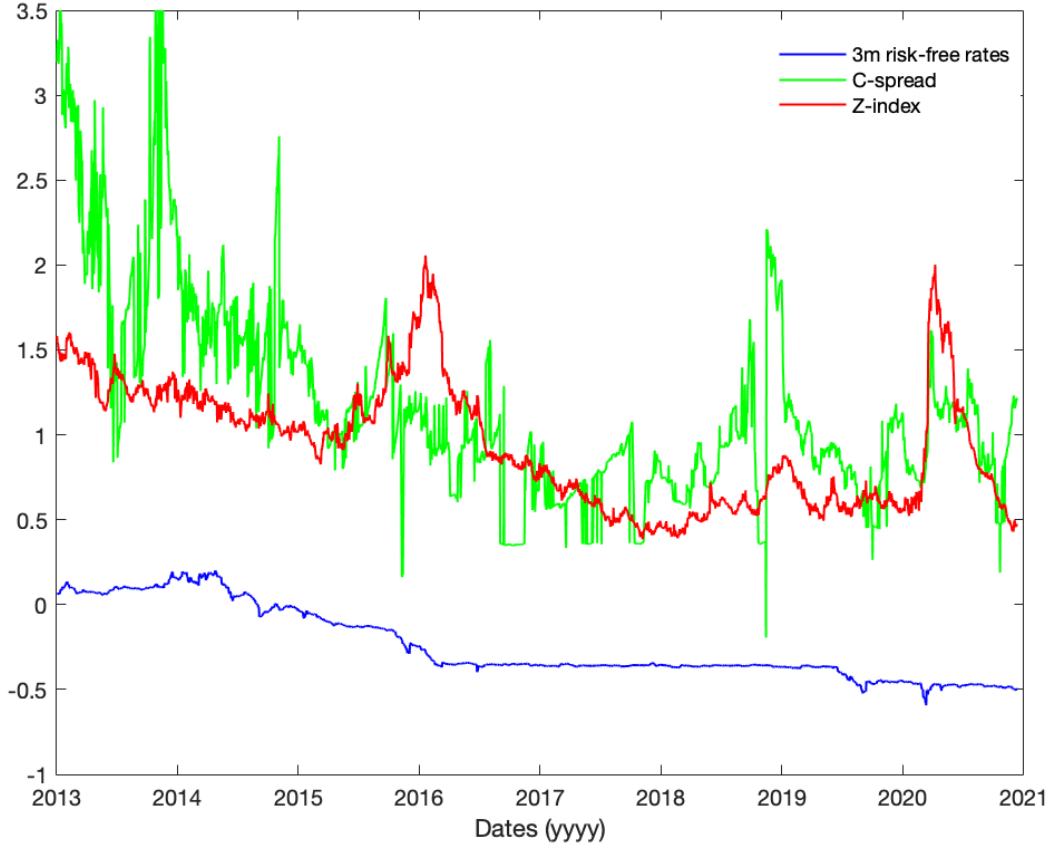


Figure 8: Three months rate r_t , C-spread C_t and Z-index Z_t for Phase III in percentage

As we can notice from the plot, C_t and Z_t have the same order of magnitude and tend to present the same behaviour on the long term, while r_t appears to be much lower and quite more constant throughout the entirety of the considered period. In order to better analyze these three time-series, we computed their mean, standard deviation and interquantile range.

	Average	Standard deviation	IQR
C_t	1.19%	0.62%	0.67%
Z_t	0.95%	0.36%	0.59%
r_t	-0.24%	0.21%	0.33%

Table 1: Descriptive statistics of the three time-series

It's interesting to notice the trend of the 3m risk-free rate: until the first half of 2014 it has been positive and around 0.1% , then it slowly lied down at -0.35% without big changes until the end of 2020. Phase III has been a period during which the ECB kept interest rates very low, even negative for most of the time, representing an expanding monetary policy, typical of periods characterized by low economic growth. Moreover, looking at the plot we can clearly see the impact of COVID-19 pandemic's outbreak (March 2020): r_t presents a small descent while both the C-spread and the Z-index have a significant and sudden upward peak, as expected during an unstable market period.

At this point we started an econometric analysis of the time-series in order to understand if C_t and Z_t are indeed related.

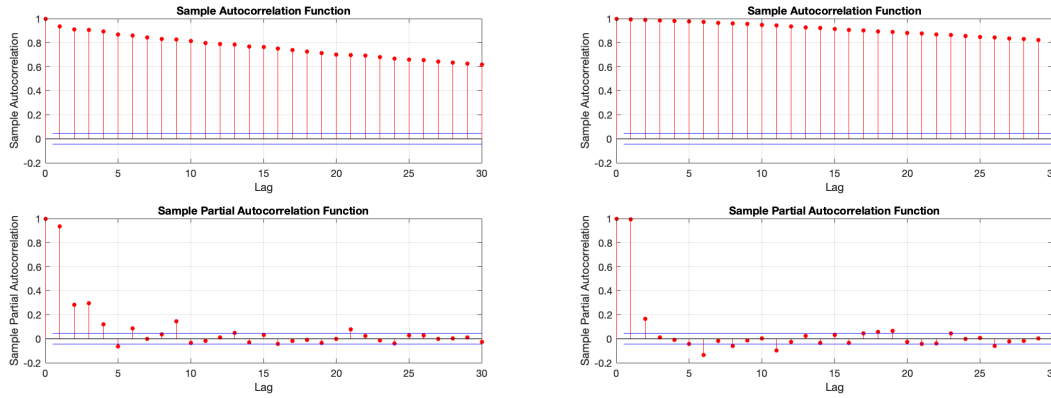
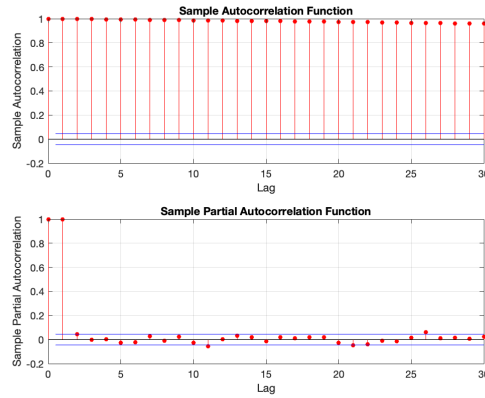
(a) ACF and PACF of C_t (b) ACF and PACF of Z_t (c) ACF and PACF of r_t

Figure 9: Correlograms of the three time-series

Looking at the Partial Autocorrelation Functions of the three time-series it seems quite clear that all of them are integrated at least of order 1 in the whole Phase III, as for both zero and one lags we can observe partial autocorrelation values significantly higher than the others.

Let's recall that an univariate time-series y_t is integrated if it can be brought to stationarity through differencing and the number of differences required to achieve stationarity is called order

of integration.

Starting from the information obtained through ACFs and PACFs we wanted to verify the non-stationarity. We firstly resorted to the KPSS (Kwiatkowski–Phillips–Schmidt–Shin), which is built in this way:

1. H_0 : data derive from a stationary process, or from a process which is stationary around a deterministic trend
2. H_1 : data derive from a non-stationary process

	p-value
C_t	1%
ΔC_t	10%
Z_t	1%
ΔZ_t	10%
r_t	1%
Δr_t	10%

Table 2: Results of the KPSS test

The KPSS test applied to our data suggests that we can refuse the null hypothesis for the time-series, but not for their first differences: we have statistical evidence to say that the three time-series have an unit root.

In order to perform this kind of analysis, it's possible to use an ADF-GLS (Augmented Dickey–Fuller) test as well, which basically experiments the opposite null hypothesis with respect to KPSS. Let's recall that ADF-GLS is a particularly powerful unit root test. It first cleans the data by removing any overall means and straightening out any linear trends. This "de-trending" makes the data more suitable for a standard unit root test (like the ADF test) to determine whether the data is integrated or not. Overall, this procedure helps identify trends in data that might not be considered by other tests.

	p-value
C_t	49.05%
ΔC_t	1.16%
Z_t	40.82%
ΔZ_t	8.24%
r_t	95.86%
Δr_t	0.00%

Table 3: Results of the ADF-GLS test

As we can see the stationarity results are consistent with the previous ones.

7 Johansen procedure for cointegration analysis

Let's start by recalling some useful concepts. An n -dimensional time-series y_t is said to be cointegrated if some linear combination $\beta_1 y_{1t} + \dots + \beta_n y_{nt}$ of the component variables is stationary. The eventual combination is called cointegrating relation, and the coefficients $\beta = (\beta_1, \dots, \beta_n)$ form a cointegration vector. There exist different procedures for testing cointegration of a multivariate time-series, in particular we preferred Johansen test over Engle-Granger's one since it permits more than one cointegrating relationship, so it's more generally applicable. Specifically, Johansen test (with both trace and eigenvalue) offers a method for identifying cointegration relationships within a set of $n \geq 2$ variables integrated of order one, which is exactly our case ($n = 3$). This method consists in conducting tests in a series until the number of cointegration relationships r is assessed.

H_0	Trace statistics	p-value	90%	95%	99%
$r \leq 0$	98.47***	0.1%	21.78	24.27	29.51
$r \leq 1$	7.75	28.8%	10.47	12.32	16.36
$r \leq 2$	0.06	84.2%	2.98	4.13	6.94

Table 4: Results of the Johansen tests using trace statistics. The last three columns are the critical values for their relative significance level

H_0	Eigenvalue statistics	p-value	90%	95%	99%
$r \leq 0$	90.71***	0.1%	15.72	17.80	22.25
$r \leq 1$	7.69	19.5%	9.47	11.22	15.09
$r \leq 2$	0.06	84.2%	2.98	4.13	6.94

Table 5: Results of the Johansen tests using eigenvalue statistics. The last three columns are the critical values for their relative significance level

Looking at the displayed results, there is statistical evidence to affirm that there is one cointegration relationship ($r = 1$), as suggested by both trace and eigenvalues tests. Thanks to Matlab function `jcitest`, which we used to perform Johansen tests, we were able to estimate the cointegration vector of the multivariate time-series as well. After having normalized by the first component, we got the following vector:

C_t	1
Z_t	-1.30
r_t	-0.46

The non-stationary term in the C-spread is explained by the linear combination $1.3Z_t + 0.46r_t$. This is a very important result, since it shows that on the long run the main driver of the C-spread is effectively Z_t .



Figure 10: $C_t - 1.3Z_t - 0.46r_t$ in percentage

As expected, this newly obtained time-series evolves around 0 and is more stable. Doing an ADF-GLS, we get a p-value equal to 0.094: there is evidence that the cointegration relationship is stationary at a 10% confidence level.

8 ECM: Error Correction Model

Regarding this section, first of all we estimated the instantaneous variance of the EU-ETS with a GARCH(1,1) model. As already done previously, we used the daily futures as proxy for the spot prices, then we computed their log-returns in order to build the GARCH model. Let's rapidly recall that the GARCH model is a generalization of ARCH (Autoregressive conditionally heteroscedastic) which allows the conditional variance to be dependent upon previous own lags. In the simplest case of a GARCH(1,1) model, which is also our case, $\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2$.

We can notice from the plot below that the log-returns are "contained" inside the standard deviation quite a few times. Moreover, every time there is a significant change in the index, the estimated value of σ_t increases immediately afterward, then gradually decreasing until the next shock.

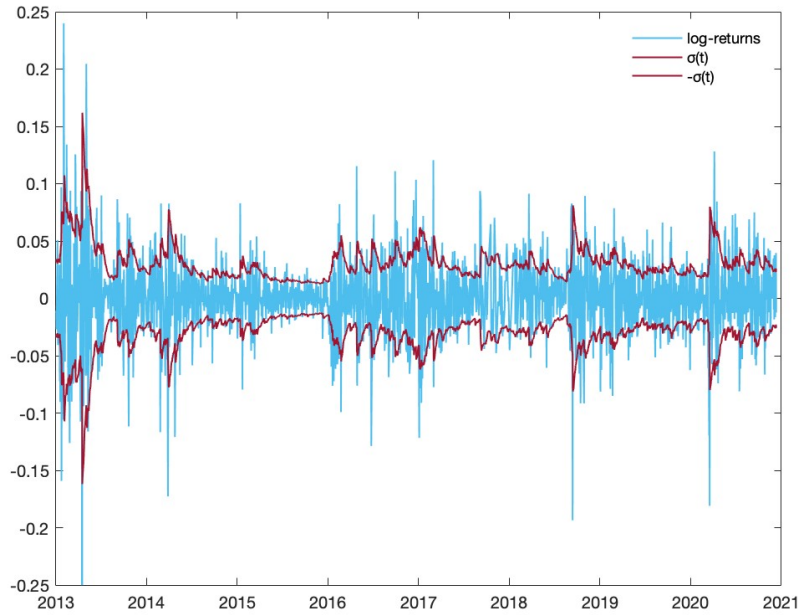


Figure 11: Daily futures' log-returns and $\pm\sigma_t$ estimated via GARCH(1,1)

Successively, following Azzone, Baviera, and Manzoni (2024), we estimated an Error Correction Model of this form

$$\Delta C_t = \alpha_0 + \sum_{i=1}^3 \beta_i \Delta C_{t-i} + \alpha_1 \Delta Z_t + \alpha_2 \Delta r_t + \alpha_3 \psi t - 1 + \sum_{j=1}^4 \delta_j Y_{j,t} + \epsilon_t$$

Where:

- In $\sum_{i=1}^3 \beta_i \Delta C_{t-i}$ we considered the first three lags of the one time differenced C-spread, since

as we can see in 9a, the first 3 lags in the PACF of C_t are statistically different from zero.

- ψ_{t-1} is the cointegration relationship found in the previous section.
- $Y_{j,t}$ are the extra variables that we considered in our analysis: SPX, VIX and WTI indexes' daily returns and previously estimated conditional volatility σ_t of the EUA spot returns.¹

	I	II	III	IV	V	VI
const	3.925e-06	7.146e-06	-9.805e-06	7.508e-06	3.496e-04	7.431e-05
ΔC_{t-1}	-0.414***	-0.0411***	-	-0.411***	-	-0.411***
ΔC_{t-2}	-0.341***	-0.339***	-	-0.339***	-	-0.338***
ΔC_{t-3}	-0.117***	-0.116***	-	-0.116***	-	-0.116***
ΔZ_t	-	0.4188***	-	0.426***	-	0.442***
Δr_t	-	0.089	-	0.100	-	0.118
Ψ_{t-1}	-0.040***	-0.041***	-	-0.041***	-	-0.041***
WTI_t	-	-	5.535e-04	6.065e-04	4.668e-04	3.502e-04
SPX_t	-	-	-	-	-8.826e-04	0.002
VIX_t	-	-	-	-	-3.423e-04	-1.032e-04
σ_t	-	-	-	-	-4.945e-04	0.001
Obs	1969	1969	1969	1969	1969	1969
BIC	-18973	-18968	-18550	-18961	-18528	-18393
AIC	-18995	-19002	-18556	-19000	-18550	-18994

Table 6: Results of 6 ECMs with different combinations of covariates. *, **, *** indicate statistical significance of the parameters at the 10%, 5%, 1% significance levels respectively

We notice that the three lags of ΔC_t , the cointegration relationship Ψ_{t-1} and the differenced Z-index ΔZ_t are always significant at 1%. Moreover, Model I is the best in terms of BIC and very close to the best in terms of AIC: we will consider exclusively this model in next robustness checks.

¹Notice that, as widely used in finance literature, we consider log-returns for WTI and SPX, while plain returns for VIX.

9 Robustness checks

The aim of this section is to repeat the analysis on the relations among the time-series we have just completed, introducing one by one different robustness tests.

In table 7 you can find the results for all of the robustness checks we did, all of them are for the model I of table 6, since we have established it to be the best model in terms of BIC.

	Phase III + IV	Open Interest	One Month	One Week
const	1.848e-05	5.893e-04	4.227e-06	7.132e-05
ΔC_{t-1}	-0.399***	-0.209***	-0.357***	-0.186***
ΔC_{t-2}	-0.328***	-0.328***	-0.337***	-0.296***
ΔC_{t-3}	-0.108***	-0.169***	-0.158***	-0.170***
Ψ_{t-1}	-0.035***	-0.104***	-0.040***	-0.126***
Obs	2449	1967	1973	1966
BIC	-24046	-16836	-18996	-16547
AIC	-24070	-16859	-19018	-16569

Table 7: ECM I under different Robustness checks

9.a EWMA Volatility

The first parameter we changed was the volatility, instead of the GARCH(1,1) we considered an EWMA volatility estimate with parameter $\lambda = 0.95$. The procedure is similar to the one we previously presented for the GARCH(1,1) model; after deriving the series of log-returns we got the variances as follows:

$$Var(t_{i+1}) = \lambda \cdot Var(t_i) + (1 - \lambda) \cdot L(t_i)^2$$

where L is the series of log-returns and, as for the variance in t_0 , we used the variance of the entire series of log-returns.

In order to compare this variances with the ones valued with a GARCH(1,1) model, we plotted them:

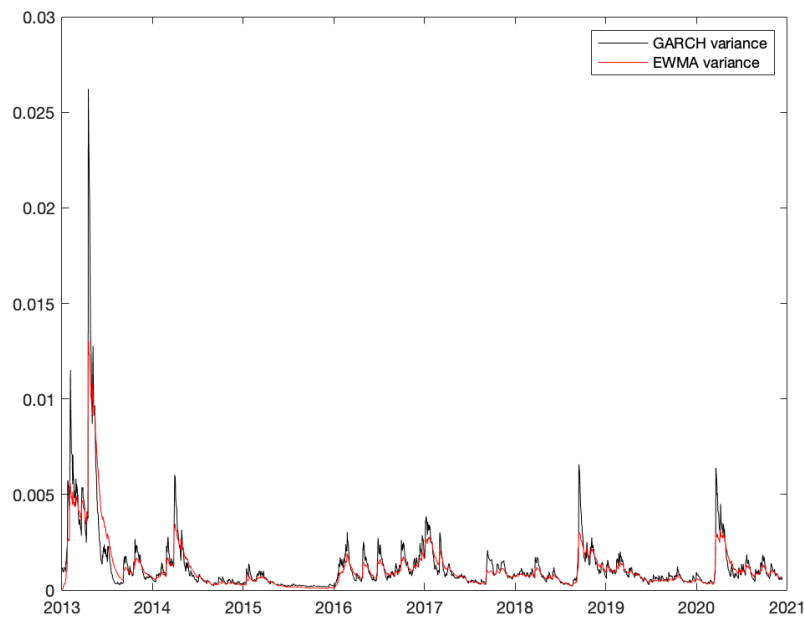


Figure 12: Volatilities with GARCH(1,1) and EWMA ($\lambda = 0.95$)

The two volatilities seem to be very similar to each other.

What are the advantages in using GARCH(1,1) over an EWMA model?

Despite EWMA being a quite simple model, thus faster and easier to implement, it associates a decreasing weight to past observations, while a GARCH model better captures volatility persistence, meaning it can better reflect the impact of past events on future volatility. Moreover EWMA's inability to capture the leverage effect might cause an underestimation of volatility in periods characterized by high yields: from the plot it's evident that when the black curve has high peaks, the red one is always a little bit lower.

We then proceeded to estimate an Error Correction Model in the same way we did in the previous section, obtaining the following results:

	V	VI
const	3.574e-04	8.583e-05
ΔC_{t-1}	-	-0.411***
ΔC_{t-2}	-	-0.3388***
ΔC_{t-3}	-	-0.115***
ΔZ_t	-	0.442***
Δr_t	-	0.119
Ψ_{t-1}	-	-0.041***
WTI_t	4.705e-04	3.484e-04
SPX_t	-8.790e-04	0.002e
VIX_t	-3.425e-04	-1.045e-04
σ_t	-7.612e-04	8.416e-04
Obs	1969	1969
BIC	-18528	-18939
AIC	-18550	-18994

Table 8: Results for the Error Correction Models using EWMA volatilities

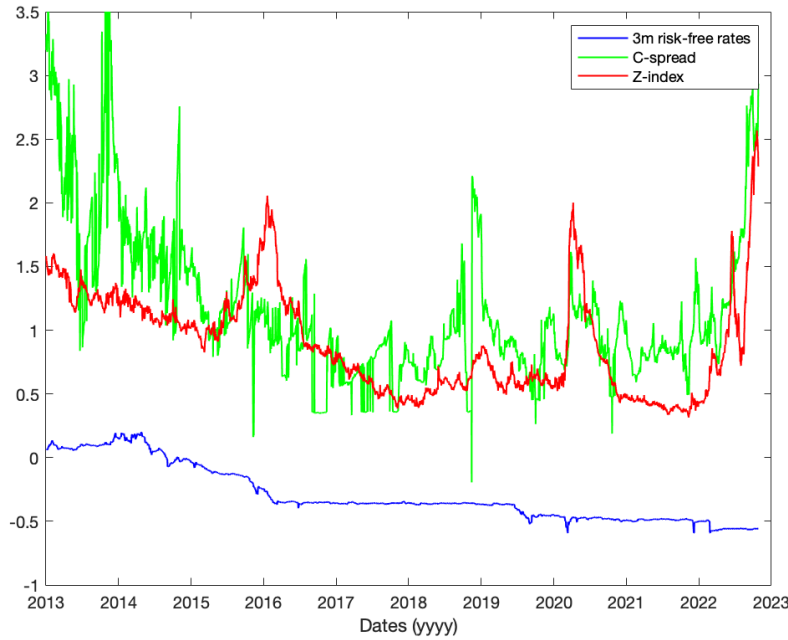
It is important to notice that we only verified V and VI models since they were the only ones which included the volatility as a covariate. As expected, both the coefficients and their significance are very similar to the one obtained in ECM that consider the GARCH(1,1) model for the volatilities. The ECM therefore is not affected by the choice of the model used to estimate the volatilities, this is connected to the impact that this covariate has on ΔC . We can see that in both models the volatilities are not really significant since they have a very high p-value (80%), this means that if we apply a slight change to them, the model will not be affected.

9.b Phase III and Phase IV

All of the discussions we have done so far have considered only the Phase III time interval, lasting until the end of 2020. We are now going to include also Phase IV (2021-2030), phase during which the linear decline of the cap has been increased to 2.2%. In our analysis however we will be stopping at the end of October 2022, since all the available data are up to that date.

We repeated all the computations with this new time horizon. The only issue we had to face regarded the delivery date of 2022: since our data end in October '22, we did not have the delivery date for the December future contracts. We therefore decided to take the same delivery day as the previous year. This choice, although it is arbitrary, still seems to be a correct approximation of the reality.

We obtained the following C-spread, Z-index and 3m risk-free rates.

Figure 13: C_t , Z_t and r_t during Phase III and IV

Something worth to notice is the behaviour the C-spread and the Z-index seem to have from the beginning of 2022: they both have a high peak which is a direct consequence of the political and energetic situation of that period: the Ukraine War began, leading to a situation of uncertainty in the Energy world and amongst companies that produced carbon, being Russia one of the world's largest producers of oil and natural gas. As a consequence, both the C-spread and the Z-index increased.

After noticing this behaviour, we wanted to verify whether the bond prices were actually effected, causing the rise of Z_t . We were expecting to experience a decrease in the bond price, synonym of "market fear", since we are working with companies that were extremely effected by the geo-politic situation. We chose a bond from ENEL ("XS1550149204") and plotted its price overtime (we chose this one as an example, mainly because the bond is valid throughout all phase III and IV).



Figure 14: Price of bond XS1550149204 issued by ENEL

As we expected, we see a decrease in the prices of the bond in correspondence with the start of the Ukraine War, with prices decreasing slowly but consistently until the end of the period. This behaviour is connected to the fact that the war is still active, therefore the uncertainty is still present.

We can also notice another time period in which the price decreases drastically (that we managed to explain historically), and it is related to COVID-19, as we have previously stated.

We then computed the same analysis as we did for Phase III only. We showed, by implying ADF-GLS test and the KPSS test, that the C-spread, Z-index and zero rates are non stationary whereas their differences are. We also computed the Johansen test in order to find the cointegration coefficients and we obtained the following relation:

$$C_t = 1.2596Z_t + 0.0698r_t$$

Something worth to mention is that the cointegration coefficient associated to the zero-rates gets even smaller than before. We justify this behaviour by looking at figure 13, towards the end the C-spread and the Z-index both drastically increase whereas the risk-free rates seem to decrease, in this way the relationship between the C-spread and the Z-index gets even stronger, thus once again strongly indicating that the credit spread is in fact the main driver of the C-spread.

Subsequently, we computed the estimation of the Error Correction Model using the GARCH(1,1) model in order to simulate volatilities. As already stated, we are going to analyze only one out of the six combinations of the covariates, model I, as it is the best model found in 8 and compare the

new results with it.

(Results can be found in table 7)

After a comparison between the coefficients of the two models, Phase III and Phase III+IV, it is easy to see that they are very similar, both in terms of value of the coefficient but also of their significance. The main difference stands in the BIC and AIC: they appear to be a lot smaller once you add Phase IV, this makes the model better and more precise. It is clearly connected to the fact that by adding a set of dates we are increasing the number of observations.

9.c ROLL-OVER OPEN INTEREST

We now want to analyze in which ways changing the roll-over rule affects the results we previously obtained. The first alternative method we tried consisted in switching, between Front and Next, on the day the open interest of the Next future is above the open interest of the Front future for the first time in the year. We obtained the following roll-over dates:

16/12/2013	01/10/2014	04/12/2015	15/12/2016
11/12/2017	23/11/2018	11/11/2019	20/11/2020

There seem to be some dates that are really close to the delivery date, if not exactly the same as the delivery date (if there was not a date in which the open interest of the Front was above the Front future we took as the roll-over date the delivery), and this may cause the C-spread to have some peaks since the time to maturity gets smaller.

After computing the C-spread and the Z-index we plotted them as we have done in section 6:

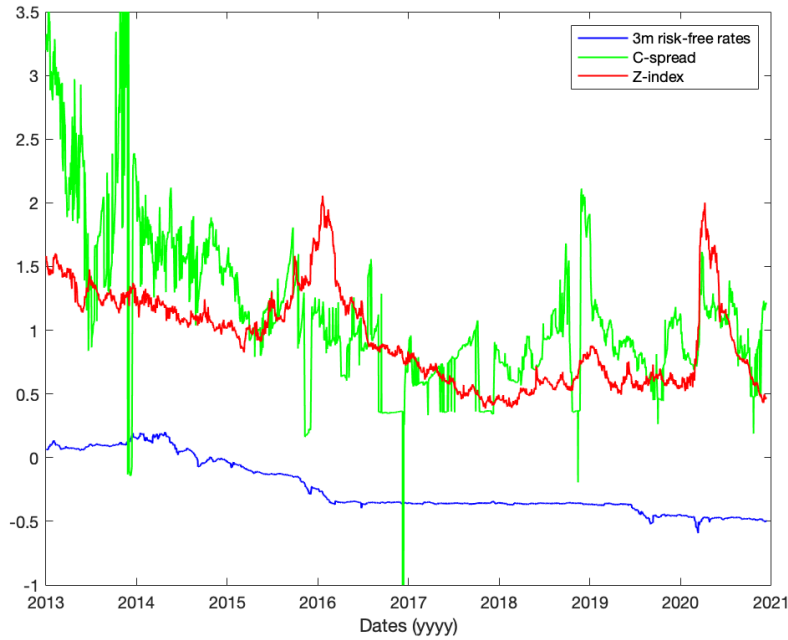


Figure 15: C_t , Z_t and r_t during Phase III with Open Interest Roll-Over

We can clearly notice two downfalls corresponding to the years in which we had to take the delivery date as the switch date. This behaviour could potentially affect the other results, we will now establish whether it happens or not.

Both the ADF-GLS and Johansen test give the same results as in the standard case with a correlation of the form:

$$C_t = 1.2594Z_t + 0.5042r_t$$

Compared with the results in section 7 the coefficient for the Z-spread is slightly smaller, whereas the one for the zero rate is a bit larger, this is possibly a consequence of the peaks we discovered earlier.

We then proceeded to estimate the ECM and the results are reported in table 7. The coefficients appear to be quite similar, with the only exception of ΔC_{t-1} that has a coefficient smaller than expected. The other big difference is the value of BIC and AIC, as they are greater than the original ones. This is probably caused by the fact that the C-spread time-series is more irregular and therefore harder to fit.

9.d ROLL-OVER one month

The second approach we tried regarding the Roll-Over rule was to switch between Front and Next exactly one month before the delivery date. We expect this choice to give very similar results compared to the ones in which we switch at the 15th of November, indeed the delivery date is usually around the middle of December, therefore going back one month is really similar to the original approach.

Everything remains practically the same, starting from the stationarity analysis up to the cointegration coefficients given from the Johansen test.

The last step we did was to estimate the Error Correction Model for model I. The results, always in table 7, are practically the same as the base case. We already anticipated this behaviour: this new roll-over rule does not differ from the original one in a tangible way. Since the difference stands in only a few dates, therefore only few points of the time-series, it does not affect the overall results.

9.e ROLL-OVER one week

The final case we analyzed is the one in which the switch, between the Front and the Next, happens exactly one week before the delivery date. This choice of Roll-Over rule might cause some peaks towards the switch date, as the time interval becomes smaller as we are approaching the delivery date, and since we are switching only one week before, we might get some strange values.

We did the usual computations and obtained the following results:

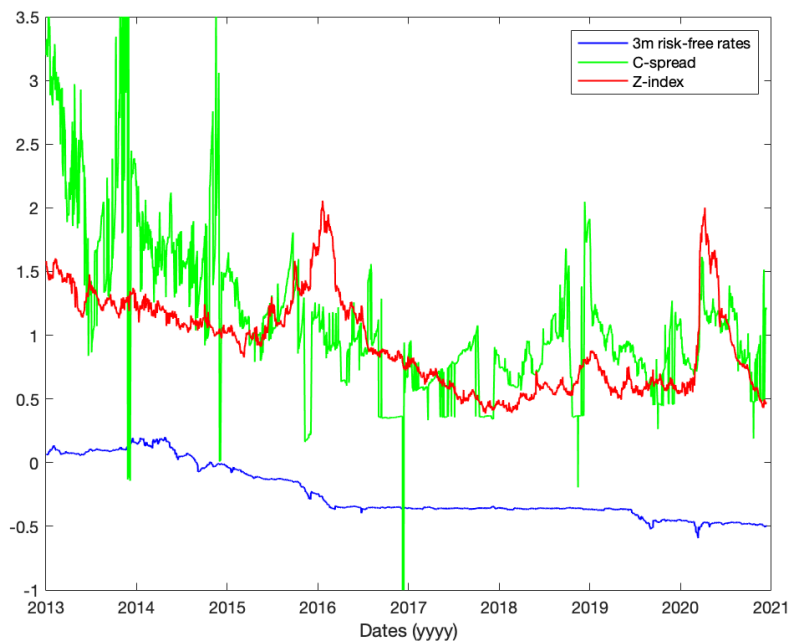


Figure 16: C_t , Z_t and r_t during Phase III with One Week Roll-Over

As expected, the plot presents a lot of peaks and irregularities as the dates get closer to the delivery dates, while these peaks were eliminated when the switch was done further away from the delivery date.

We then computed the ADF-GLS test and obtained the same results as the base case. The Johansen test returned the following cointegration coefficients:

$$C_t = 1.2793Z_t + 0.5943r_t$$

In this case the coefficient connected to the risk-free rates becomes bigger, meaning that r_t becomes more significant in explaining the C-spread. This behaviour is probably connected to the peaks caused by the change of roll-over rule.

To end the analysis, we estimated the ECM for model I and we obtained very similar results to the case in which we used the open interest as the reference for the switch. Both the coefficients and BIC and AIC are very similar, even though the two series are not exactly alike. Indeed, the two series both present a bigger amount of peaks, even if they are explained by different causes. Nevertheless this peaks result in similar effects, yielding an estimate not as precise as the base case one.

At the end of these robustness checks we can affirm that the econometrics results are in fact consistent: the main assumptions of our analysis do not significantly interfere with its results.

10 Quantile Regression

We were asked to estimate quantile regressions on models (I) and (VI), previously presented in section 8.

With this aim in mind, we had to estimate, for quantiles values ranging between 0.1 and 0.9, the coefficients β of the independent variables, by solving

$$\beta_q = \min_{\beta_q} \left\{ \sum_{i=1}^N \rho_q(Y_i - x_i \beta_q) \right\}$$

with $\rho_q(r) = r(q - I(r < 0))$ being the check function for the different quantiles q .

In order to compute this quantile regression, we implemented `quantileRegression`, which outputted the coefficients β , the bootstrapped standard error, as justified in Goncalves and H. White (2005), and the significance p-values computed with the t-statistic.

We organized our results in two tables, one for each model, as done in Palao and Pardo (2021)

	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
ΔC_{t-1}	-0.269*** (0.034)	-0.108*** (0.014)	-0.067*** (0.006)	-0.060*** (0.003)	-0.056*** (0.002)	-0.060*** (0.003)	-0.078*** (0.005)	-0.114*** (0.010)	-0.280*** (0.033)
ΔC_{t-2}	-0.177*** (0.050)	-0.079*** (0.014)	-0.040*** (0.004)	-0.036*** (0.004)	-0.033*** (0.004)	-0.032*** (0.004)	-0.043*** (0.006)	-0.074*** (0.009)	-0.179*** (0.027)
ΔC_{t-3}	-0.043 (0.035)	-0.022 (0.014)	-0.018*** (0.004)	-0.019*** (0.002)	-0.009 (0.004)	-0.016 (0.004)	-0.020 (0.004)	-0.037*** (0.010)	-0.099*** (0.025)
Ψ_{t-1}	-0.074*** (0.017)	-0.038*** (0.007)	-0.016*** (0.002)	-0.005*** (0.001)	-0.003** (0.001)	-0.001 (0.001)	0.001 (0.000)	0.009* (0.005)	0.006 (0.007)
const	-0.002*** (0.001)	-0.000*** (0.000)	-0.000*** (0.000)	0.000 (0.000)	0.000** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.001*** (0.000)

Table 9: Results of quantile regrssion analysis for model (I), as presented in 6. Figures in brackets denote the standard error. The ***, ** and * indicate rejection of the null hypothesis at the 1%, 5% and 10% level, respectively

By analyzing these results, we can see that the values of the coefficients seem coherent with the ones we found in section 8. The main difference to notice is that in some cases the results seem to indicate the constant's signficancy (while still having a coefficient that is practically null), whereas previously we had always found its coefficient to be not significant. We were not expecting this behavior, as the coefficients are in fact all of order lesser than 10^{-3} .

We then repeated the same procedure for model VI, which contains all the possible covariates.

	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
ΔC_{t-1}	-0.257*** (0.034)	-0.109*** (0.014)	-0.066*** (0.006)	-0.060*** (0.003)	-0.056*** (0.003)	-0.060*** (0.003)	-0.078*** (0.005)	-0.114*** (0.010)	-0.280*** (0.033)
ΔC_{t-2}	-0.177*** (0.050)	-0.079*** (0.014)	-0.040*** (0.004)	-0.036*** (0.004)	-0.034*** (0.004)	-0.032*** (0.004)	-0.043*** (0.006)	-0.074*** (0.009)	-0.179*** (0.027)
ΔC_{t-3}	-0.043 (0.035)	-0.022 (0.014)	-0.018*** (0.004)	-0.019*** (0.002)	-0.009** (0.004)	-0.016*** (0.004)	-0.020*** (0.004)	-0.037*** (0.010)	-0.099*** (0.025)
ΔZ_t	0.362 (0.237)	0.146 (0.094)	0.080** (0.028)	0.047*** (0.016)	0.058*** (0.017)	0.023 (0.020)	0.095*** (0.025)	0.183*** (0.057)	0.438** (0.187)
Δr_t	0.677 (0.498)	0.152 (0.343)	0.093 (0.183)	-0.870*** (0.102)	-0.077 (0.081)	-1.039*** (0.163)	0.317* (0.157)	0.337 (0.249)	0.352 (0.746)
Ψ_{t-1}	-0.066*** (0.017)	-0.036*** (0.006)	-0.016*** (0.002)	-0.008*** (0.001)	-0.004*** (0.001)	-0.001 (0.001)	-0.000 (0.000)	0.007 (0.005)	-0.003 (0.007)
WTI_t	-0.003 (0.002)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.001*** (0.000)	0.000 (0.000)	-0.001 (0.000)	0.001 (0.001)	0.001 (0.002)
SPX_t	0.006 (0.006)	-0.001 (0.002)	0.000 (0.000)	0.005*** (0.001)	-0.003*** (0.001)	-0.002** (0.001)	-0.000 (0.001)	-0.002 (0.002)	-0.003 (0.005)
VIX_t	-0.001 (0.001)	-0.000*** (0.000)	-0.000** (0.000)	0.000*** (0.000)	-0.000*** (0.000)	-0.000** (0.000)	-0.000* (0.000)	0.000 (0.000)	0.000 (0.000)
σ_t	0.002 (0.002)	0.001 (0.002)	0.001 (0.002)	0.009** (0.003)	0.002 (0.002)	0.002 (0.002)	0.002 (0.004)	0.001 (0.001)	0.001 (0.003)
const	-0.001 (0.001)	0.000 (0.000)	0.000* (0.000)	-0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.001*** (0.000)

Table 10: Results of quantile regression analysis for model (VI), as presented in 6. Figures in brackets denote the standard error. The ***, ** and * indicate rejection of the null hypothesis at the 1%, 5% and 10% level, respectively

These results seem to be more aligned with the one we found using the basic linear regression model: the lagged Delta C-spread seem, less as the lag increases, relevant for almost all quantiles, and the same can be said for the σ_t and the Ψ_t .

The coefficients of the constant term are almost null, even if the p-values seem to indicate, for some quantile, their relevance for the regression. There also seems to be a switch in significance between the ΔZ_t and the Δr_t for a couple of quantiles, where the risk-free rate appears to be more significant than the Z spread. For a few lower quantiles some of the Y variables (such as the SPX_t and the VIX_t) gain relevance, but as can be seen the coefficients have still very low values.

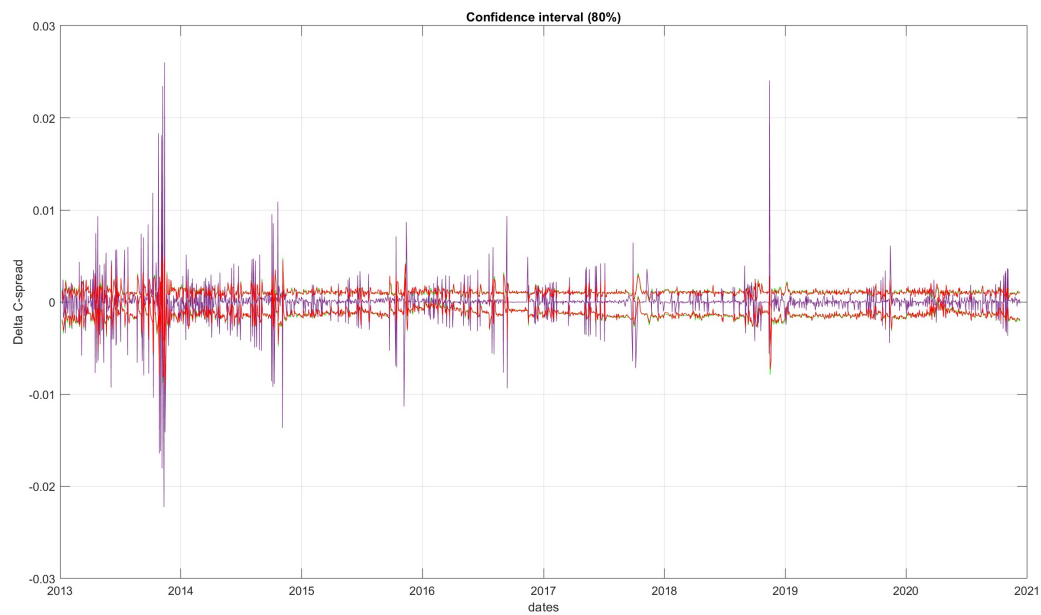


Figure 17: Confidence Interval at 80% for both model (I) in green and (VI) in red

This plot shows the real time-series compared to the predicted ones with quantile 80%. As we can see the predicted series contains the real one nicely, exception made for the peaks in the series that the regression is not really able to capture. We have repeated this plot for all of the quantiles we considered: the main thing we have observed is that as the quantile gets smaller, therefore the range of percentile becomes slimmer, the prediction worsen and is not able to replicate the model in a correct way.

Graphically model I and VI give almost the same results, as can be seen in Figure 17, where the estimates of the two models overlap almost completely.

11 EUA synthetic repo

Let's start by trying to clarify what a *synthetic repo* is, comparing it with a traditional repo (Caceis (2010)).

A traditional repo:

1. Involves the temporary sale of a security with the agreement of a later repurchase at a fixed price.
2. The security itself works as collateral.
3. Is often used as a short-term financing procedure.

A synthetic repo:

1. Replicates the economic meaning of a traditional repo without the exchange of an underlying security.
2. A collateral is not present.
3. Is especially used in markets where repo is limited to a few selected participants or where a repo structure has not been fully developed.

A market player involved in the EU-ETS could potentially finance itself via an EUA synthetic repo:

- short position on a spot EUA, thus obtaining funds at start time t
- long position on an EUA future, paying back the loan at T

This would allow the corporate to borrow funds only utilizing EUA spot and future, at an interest rate equivalent to the sum of zero-rate and C-spread, as

$$F_{t,T} = S_t e^{(r_{t,T} + C_t)(T-t)}$$

Having previously in the paper shown the correlation between C-spread and the rates, represented by the Z-index, at which the market player usually gets financing, we can affirm that the two shouldn't show significant differences in general.

It would anyway be important to consider the future contract maturity, as the liquidity of the contract in said maturity could heavily affect its price and therefore the financing rate.

Other issues would be the market risk as well as the counterparty risk, as there could be significant price fluctuations in the EUA futures market, or one of the counterparties could not be able to honor its duties.

Let's add that a player of EU-ETS should prefer financing herself through synthetic repos, instead of traditional methods, if her credit spread is bigger than the C-spread. In figure 8 the Z-index is often smaller than C_t , but we recall that Z_t is an average over the biggest companies involved in the market, thus it can happen, especially for a smaller firm (less reliable), that her credit spread is bigger than the C-spread, pushing her to get funds via synthetic repos.

In Azzone, Baviera, and Manzoni (2024), the authors, having identified this market inefficiency caused by the constantly positive C-spread, claim that the most effective solution would be to include the EUA in the list of Euro-system eligible collateral, allowing corporates involved in the EU-ETS to access financing at the ECB policy rate.

The formation of an active EUA repo market would eliminate the C-spread anomaly, driven by the current financing rates, increasing EU-ETS market efficiency and positively affecting the green transition, helping European Union to reach the medium/long term emission targets.

12 Matlab Code

The purpose of this section is to give some information about the code.

- In order to compute the ADF-GLS, since there wasn't an already implemented Matlab function, we imported the function from Python using Matlab's `py.` functionality. The code runs smoothly, but since it needs the Python `arch` library to run, to obtain the results it has to be activated setting the stationarity flag, in our `stationarity_tests` function, to 1.
- In order to compute the robustness checks in section 9, we implemented the code in a way that by changing the flags at the beginning, you can choose the settings you want.
- At line 270 starts a section where different covariates for the ECM can be chosen.

Finally we report the summary of the computational time of our code: we paid close attention to the computational time of the `Z_spread` function, as it is certainly the most computationally expensive one. We tried to make it as fast as possible by maximizing vectorized operations at the expense of cyclic ones.

Function Name	Calls	Total Time (s)	Self Time* (s)	Total Time Plot (self time / total time)
Run_FinalProject_Group3A	1	88.942	4.467	
Z_spread	12	45.233	18.768	
interp1	433517	13.306	10.738	
readtable	156	11.395	0.149	
fzero	224848	10.186	6.449	
OIS_preprocessing	1	8.614	0.380	
ExecutableFunction>ExecutableFunction.validateAndExecute	156	8.479	0.044	
ReadTable>ReadTable.execute	156	6.950	0.083	
variance_estimation	1	4.791	0.272	
DetectImportOptions>DetectImportOptions.execute	156	4.216	0.012	
DetectImportOptionsText>DetectImportOptionsText.execute	156	4.195	0.025	
DetectImportOptionsText>DetectImportOptionsText.getTextOpts	156	4.115	0.012	
garch.estimate	1	3.944	0.239	
Z_spread>@(z)NPV*exp(-z*delta(k,:))-bonds_prices(k,jj)/100	1845048	3.737	3.737	
boxplot	2	3.733	0.132	
yearfrac	2843	3.095	0.503	
Cspread	31	3.010	0.132	
autocorr	5	2.817	0.646	
FunctionStore>FunctionStore.getFunctionByName	156	2.756	0.002	
FunctionStore>accessMap	156	2.754	2.032	
newplot	37	2.515	0.652	
prepareAxes	29	2.383	0.049	

Figure 18: Computational time summary

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